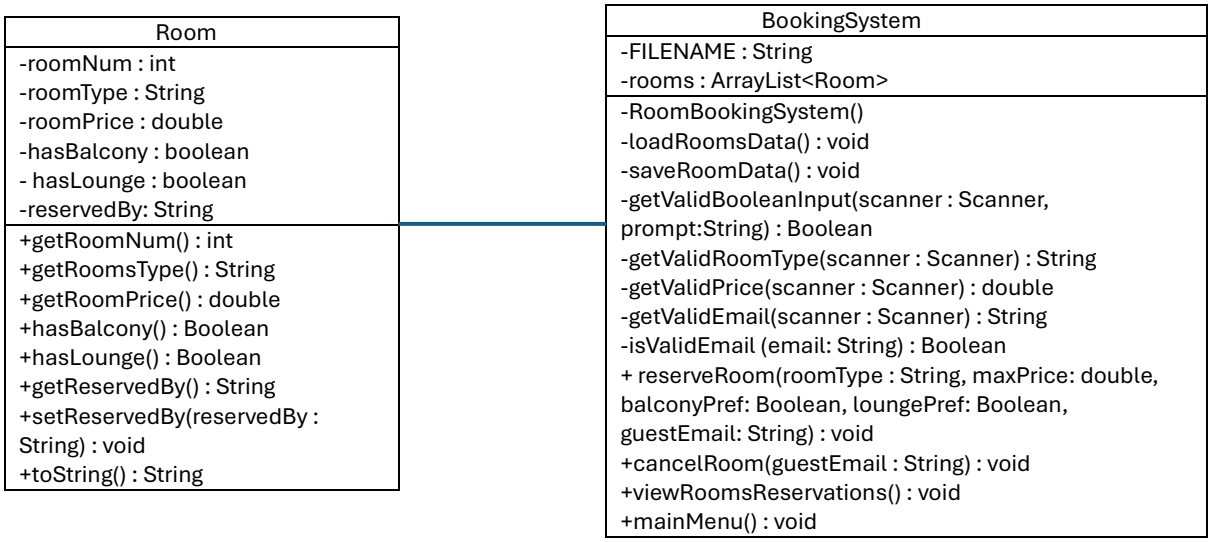
CourseWork 2 : Report

The Room Booking System is a Java software application that is aimed at controlling room reservations in an efficient manner by making it easy for users to operate the system. It enables guests to book or even cancel rooms depending on the kind of room they prefer, its price range and other features such as lounge or balcony. The system has been designed with the use of object oriented programing principles in order to make it easy to manage and expand. For ensuring that the room data remains consistent across different sessions, an external file named rooms.txt is used, that is opened whenever the application is launched and modified after each and every session.

Another major concern that was emphasized during the course of this project, was the effectiveness of the data validation that was implemented, so as to minimize the chances of making mistakes and even to protect the integrity of the booking process. In particular, the projects focus area was on how to verify emails for online signup and registration. Because email inputs are used as invariant for reservations and cancellations, it was very important to safeguard its accuracy. After researching various methods, I settled on a solution that involved the use of regular expressions. The isValidEmail() function compares the input from users with a normative template structure of email wording to correct deviations of its anatomy language. For example, a statement such as invalidemail.com would be put out of consideration. Such an approach also improved the overall consistency of the system, while offering a practical and modular mechanism for input validation. The exploration of Java's Pattern and Matcher classes for regex processing was an invaluable part of this development.

The system also applies other validation methods, such as allowing a user-defined price range for room reservations to constrain values within particular limits. For example, users are asked to put a price from 0 to 60, any price which is higher or lower generates an error and the user is informed to try again. This prevents such invalid entries from affecting the working of the booking system. Moreover, the system includes a next-best match functionality that enables users to choose from alternative rooms in the event their desired rooms are unavailable, thus enhancing their experience even more.

In the case of the Room Booking System, the integration resulted from responsible integration of research, modularization, and validation is a good demonstration of use of object based programming to address assorted problems. The incorporation of functionalities such as email validation or next-best recommendations also reveal the level of design and application investment that was made to ensure that the application is reliable and easy to use.



Room booking structure system:

Room Booking System

- **List of Rooms**: Represents a collection of all available and reserved rooms.
- **Display List of Rooms**: Shows all rooms available for reservation.
- **Load Room Data**: Loads room details from a data source or file.
- **Reserve Room**: Allows a user to select and reserve a room based on their preferences.
- **Cancel Room Reservation**: Cancels a reservation for a specific room.
- **View Reserved Rooms** : Displays a list of all reserved rooms.
- **Quit** : Terminates the system.

Room

Attributes:

- Number - Unique identifier for the room.
- Type - Room type (e.g., Single, Double, Suite).
- Price - Cost of the room per night.
- Balcony - Indicates if the room includes a balcony.
- Lounge - Indicates if the room includes access to a lounge.
- Reserved By - Stores the name or email of the person who reserved the room.

Behaviour:

1. Reserve
2. Cancel

Behavioural Overview

1. **Room Booking System:**

List of Rooms	Display List of Rooms	Load Room Data
Reserve room	Cancel room reservation	View Reserved room
Quit		

2. **Room**

Number	Type	Price
Balcony	Lounge	Reserved by